

Final Project: Text-Driven Image Editing

Course: Deep Generative Models (ESE 6450-001, University of Pennsylvania)

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Text-to-image generative models can generate vivid and photo-realistic images. In this project, we will experiment with multimodal diffusion models and push the limit of their generative and creative power. An interesting and fun question that we will work on in this final project is: Given a source image, can we utilize the power of text-to-image diffusion models and vision-language models to perform image editing based on different concepts? For example, can we edit a given face image to make the face “angry” or “smiling”? We will test several baseline methods for image synthesis by combining diffusion models with the vision-language models we studied in class.

Teaming. Please form a team and fill out the names of your team members at the desired time slot in groups_teaming.xlsx. We expect each group to have 3 members. The expectation of work quality and project scale will be proportional to the team size, with larger teams held to a higher standard.

1 Project Description

This project is to be completed by a team of three students. While in exceptional circumstances teams of size two or four will be allowed,¹ note that the expectation of invested efforts, contributions, and novelty will be higher for a larger team. That is, a larger group is expected to incorporate more features in the project than smaller teams.

In this project, you will start by implementing baseline methods for diffusion-based image editing. Table 1 lists well established approaches to diffusion-based image editing, such as DDIM inversion, which provide a foundation for developing more innovative techniques. To compare baseline methods, you will use the PIE-Bench repo, a comprehensive editing benchmark designed to evaluate image editing tasks across various categories. Next, you will extend these baseline methods by developing innovative and novel approaches to image editing, improving their accuracy and edit-ability while preserving of the original image’s key attributes. You are encouraged to explore novel architectures, fine-tuning strategies, or modifications to the diffusion process itself to achieve these results.

Table 1: Summary of Key Baseline Methods.

Method	Description	Strengths
DDIM [15]	Diffusion-based inversion with prompt-to-prompt editing	Good for content-aware editing
Direct Inversion [7]	Direct image-to-latent space inversion with p2p	Quick and simple editing but less flexible
Null Text Inversion [12]	Null-text inversion with prompt-to-prompt	Good balance between retention and editability
Blended Latent Diffusion [1]	Blending source and target latents for edits	Flexible in blending elements for complex edits
Edit Friendly P2P [6]	Edit-friendly localized image manipulation	Ideal for localized precision edits
InstructDiffusion [4]	Instruction-driven text-guided diffusion	Highly tailored, instruction-based edits
InstructPix2Pix [2]	Instruction-based Pix2Pix framework	Effective in style and content transformation
MasaCtrl [3]	Control-based editing method	Precise control over edits
StyleDiffusion [9]	Focuses on changing image style	Best for aesthetic/style-based transformations
Pix2Pix Zero [13]	Zero-shot learning Pix2Pix method	Effective for zero-shot image editing tasks

Here are some milestones to assess the progress of the project. Grading details are further discussed in Section 4.

1. Evaluate at least n baseline methods in PIE-Bench, where n is the number of members in the team, and report metric results on editing effectiveness, consistency preservation, and/or other indicators relevant to your motivation.
2. Propose an improved editing method by introducing novel techniques. Improvements can be in multiple directions such as image quality, editing effectiveness, time efficiency, background preservation, additional modality control, etc.

¹For teams of two or four, please email us to ensure we have sufficiently number of teams for everyone in the class.

3. Demonstrate the final results of the developed method and compare them with baseline performance.
4. Present your project and write a report describing the method, experiments, and results.

2 Examples of Project Ideas

You are encouraged to make your innovation in multiple directions, as long as your approach and experimental results on the benchmark justify your motivation. Here we provide some potential ideas for your inspiration.

1. **Enhanced Text Encoder:** Replace the default text encoder (often CLIP or BERT) used in baseline methods with a better text encoder, such as OpenAI’s CLIP with larger multimodal models, or the latest version of a transformer-based language model such as LLaMA. This would improve the richness of the text understanding, enabling more precise and nuanced edits.
2. **Enhanced Diffusion Backbone:** Replace the diffusion model backbone (e.g. Stable Diffusion 1.4/1.5) with the more powerful Stable Diffusion 2.0/XL for higher fidelity and better resolution handling in image editing tasks.
3. **Fast and Efficient Editing:** Implement consistency models [16] or other distilled methods [18] to accelerate the diffusion process. Instead of running diffusion processes using the entire trajectory, a new type of model known as consistency models [16] can predict the output after fewer steps, reducing the computational time while maintaining the quality of image edits. Alternative, you could combine a few steps of diffusion with matching flows [10].
4. **ControlNet for More Modalities:** Apply edits under constraints using pose estimation models (e.g., OpenPose [11]) or segmentation models (e.g., Segment Anything [8, 14]) with ControlNet [17] to maintain structural consistency in human and object pose editing.
5. **Parameter-Efficient Fine-Tuning:** Collect additional *instructional* text-image datasets and use parameter-efficient fine-tuning (e.g., LoRA [5], Instruction Tuning [2]) to adapt the diffusion backbone for better editing performance.

While your team is encouraged to come up with your own idea, the above list serves as a guideline to the caliber of depth and novelty you will be graded on. If you do something different, to ensure your idea passes the bar, your team may consult the TAs in private during office hours.

3 Evaluation and Benchmarks

To ensure fair evaluation across teams, the dataset of choice for this project will be **PIE-Bench** [7], a large-scale prompt-driven image editing benchmark designed to evaluate diffusion-based editing methods. PIE-Bench features:

1. 700 images with 10 types of diverse editing tasks.
2. Detailed implementation of the baseline methods mentioned in Table 1.
3. A balanced distribution of artificial and natural images, covering categories such as animals, humans, indoor, and outdoor scenes.
4. Detailed annotations for each image, including editing prompts, blended words, and mask annotations.

PIE-Bench covers a wide range of variations in terms of object types, scene compositions, and background changes, providing ample opportunities for students to experiment with different editing techniques. The dataset is available for non-commercial research purposes and can be accessed through the following link: [data](#).

For a fair comparison, we request you to report the following metrics from the benchmark for evaluation: 1) SSIM, 2) LPIPS, and 3) the CLIP similarity on the target prompt with the target image.

4 Instruction and Deliverables

The total points for the final project is 100. The project will be graded based on the following:

1. **Presentation:** The presentation should consist of but not be limited to: an introduction to the problem, the motivation for your approach, the choice of model and the innovation, and the baseline and new results obtained.

All presentations will take place in AGH 615 on December 2nd. The room has limited capacity, and to ensure smooth presentation, we enforce the following policies:

- Within a team, each member is expected to speak for roughly 3 minutes. After 9 minutes, there will be 5 minutes in total for the instructor and TAs to ask questions.
 - Between each presentation, we leave a minute of buffer time to switch to the next team. In the final minute, we expect the following: (1) the team finishing the presentation exits the room; (2) the next team gets ready and shares the screen over zoom; (3) the next next team enters the room. This means each team needs to arrive 15 minutes earlier than the assigned time slot, and waits for the previous team to finish.
 - Each presenter is encouraged to spell out his or her name before presenting the approach and result. Or the team member needs to be introduced during the presentation. This allows us to better evaluate individual performance.
 - We allocate a score of 5 for the team which finishes the presentation on time.
2. **Final Report:** The report is limited up to eight (8) pages in the style of the CVPR 2026 Template². This does not include additional figures, references, and contributions sections. It should have a title, abstract, introduction, related work, results, methods, experimental details, discussion, future work, and bibliography. The report should also clearly indicate which team members contributed which parts to the project, and this should be roughly equally divided among the team members.
 3. **Content:** We expect a clear motivation, relevant background, explanation of contributions, and how it differs from the existing baselines. The related work should cover relevant articles such as modified version of backbone models and describe their results appropriately. The experimental setup should be clearly described and results and discussion examined logically. The goal is to show creativity in experimental setup, a thorough analysis of results or potential explanations why you think that the results were unsatisfactory, and potential improvements. The three metrics mentioned above in the Evaluation section should also be included.
 4. **Contributions:** At the end of your report, please also include a small section that clearly indicates each project member's contributions to the project. For example, suppose Adam, Becky and Grace are in a group. Adam is responsible for experiment 1 and writing the Introduction of the report. Becky is responsible for experiments 2-3 and writing the Results section of the report. Grace is responsible for experiments 4-5 and the writing of the related work and conclusion section. The goal of this section is to ensure that the amount of work is fairly distributed among team members.

²[https://github.com/cvpr-org/author-kit/archive/refs/tags/CVPR2026-v1\(latex\).zip](https://github.com/cvpr-org/author-kit/archive/refs/tags/CVPR2026-v1(latex).zip)

References

- [1] O. Avrahami, D. Lischinski, and O. Fried. Blended diffusion for text-driven editing of natural images. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 18208–18218, 2022. 1
- [2] T. Brooks, A. Holynski, and A. A. Efros. Instructpix2pix: Learning to follow image editing instructions. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 1, 2
- [3] M. Cao, X. Wang, Z. Qi, Y. Shan, X. Qie, and Y. Zheng. Masactrl: Tuning-free mutual self-attention control for consistent image synthesis and editing. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 22560–22570, 2023. 1
- [4] Z. Geng, B. Yang, T. Hang, C. Li, S. Gu, T. Zhang, J. Bao, Z. Zhang, H. Li, H. Hu, et al. Instructdiffusion: A generalist modeling interface for vision tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12709–12720, 2024. 1
- [5] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>. 2
- [6] I. Huberman-Spiegelglas, V. Kulikov, and T. Michaeli. An edit friendly DDPM noise space: Inversion and manipulations. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12469–12478, 2024. 1
- [7] X. Ju, A. Zeng, Y. Bian, S. Liu, and Q. Xu. Pnp inversion: Boosting diffusion-based editing with 3 lines of code. *International Conference on Learning Representations (ICLR)*, 2024. 1, 2
- [8] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, et al. Segment anything. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4015–4026, 2023. 2
- [9] S. Li, J. van de Weijer, T. Hu, F. S. Khan, Q. Hou, Y. Wang, and J. Yang. Stylediffusion: Prompt-embedding inversion for text-based editing. *arXiv preprint arXiv:2303.15649*, 2023. 1
- [10] Y. Lipman, R. T. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. *arXiv preprint arXiv:2210.02747*, 2022. 2
- [11] G. H. Martinez. Openpose: Whole-body pose estimation. *Ph. D. dissertation*, 2019. 2
- [12] R. Mokady, A. Hertz, K. Aberman, Y. Pritch, and D. Cohen-Or. Null-text inversion for editing real images using guided diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6038–6047, 2023. 1
- [13] G. Parmar, K. Kumar Singh, R. Zhang, Y. Li, J. Lu, and J.-Y. Zhu. Zero-shot image-to-image translation. In *ACM SIGGRAPH 2023 Conference Proceedings*, pages 1–11, 2023. 1
- [14] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. 2
- [15] J. Song, C. Meng, and S. Ermon. Denoising diffusion implicit models. *International Conference on Learning Representations (ICML)*, 2020. 1
- [16] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever. Consistency models. In *International Conference on Machine Learning*, pages 32211–32252. PMLR, 2023. 2
- [17] L. Zhang and M. Agrawala. Adding conditional control to text-to-image diffusion models. In *ICCV*, 2023. 2
- [18] Z. Zhou, D. Chen, C. Wang, C. Chen, and S. Lyu. Simple and fast distillation of diffusion models. *Advances in Neural Information Processing Systems*, 37:40831–40860, 2024. 2